

AWS Console Deployment Guide

Deploying Meridian Goods Using Only the AWS Website — Every Click Explained

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Repository: github.com/Rajshekhar6666/cloudtask-aws

READ THIS FIRST

This guide assumes you have never used AWS before. No terminal, no command line, no installing software — every single step happens by clicking things on the AWS website. Follow the steps in order and don't skip any. Every step tells you exactly what to click and exactly what you should see afterward. If what you see doesn't match, stop and check the Troubleshooting table at the end before continuing.

ONE RULE THAT MATTERS MOST

AWS shows a region name in the top-right corner of every page (e.g. "Asia Pacific (Mumbai)"). Once you pick a region in Step 1, do not change it for the rest of this guide. Your database and your server must be created in the same region, or they will not be able to find each other.

What You'll Build

A small online product catalog website, running on two AWS resources:

- An EC2 instance (a small virtual server) that runs the website
- An RDS database (a managed MySQL database) that stores the products

You'll create two security groups first (these act like firewalls, deciding what's allowed to talk to what), then the database, then the server. Total time: about 30-40 minutes, most of which is just waiting for AWS to finish creating things.

Steps at a glance

- Part A — Sign in and pick your region (Step 1)
- Part B — Create two security groups (Steps 2-3)
- Part C — Create the database (Steps 4-5)
- Part D — Prepare your startup script (Step 6)
- Part E — Launch the server (Steps 7-8)
- Part F — Check everything works (Steps 9-12)
- Part G — Take your screenshots (Step 13)
- Part H — Delete everything when you're done (Steps 14-16)

Part A — Sign In and Pick Your Region

Step 1 — Sign in and set your region

1. Go to <https://console.aws.amazon.com> in your browser and sign in with your AWS account email and password.
 2. Look at the very top-right corner of the page. You'll see a region name, such as "N. Virginia" or "Mumbai", next to a small globe or map-pin icon.
 3. Click on that region name. A dropdown list of all AWS regions appears.
 4. Choose "Asia Pacific (Mumbai) ap-south-1" (or, if you're not in India, whichever region is closest to you — just remember which one you picked).
 5. Confirm the top-right corner now shows your chosen region.
- ✓ **How you'll know it worked:** The top-right corner of every AWS page now shows the same region name, no matter which service you visit.

TIP

Keep this guide open in a window next to your browser so you can follow each click one at a time.

Part B — Create the Security Groups

A security group is a set of rules controlling what network traffic is allowed in and out of a resource. You'll create two: one for the server (allows website visitors in), one for the database (allows only the server in, nobody else).

Step 2 — Create the server's security group (meridian-ec2-sg)

1. In the search bar at the very top of the AWS Console (it says "Search" with a magnifying glass icon), type: EC2
 2. Click on "EC2" in the dropdown results — this opens the EC2 service dashboard.
 3. In the left-hand sidebar menu, scroll down to the "Network & Security" section, and click "Security Groups".
 4. Click the orange "Create security group" button in the top-right of the page.
 5. In the "Security group name" box, type: meridian-ec2-sg
 6. In the "Description" box, type: Allows web traffic and SSH for the Meridian Goods app server
 7. Under "VPC", leave it set to the default VPC (it will already be selected — look for the one marked "default").
 8. Scroll down to "Inbound rules". Click "Add rule".
 9. For this new rule: set "Type" to "HTTP" (the Port range fills in automatically as 80). Set "Source" to "Anywhere-IPv4" (this fills in 0.0.0.0/0 automatically).
 10. Click "Add rule" again to add a second rule.
 11. For this second rule: set "Type" to "SSH" (Port range fills in as 22 automatically). Click the "Source" box, and from the dropdown choose "My IP" — AWS automatically detects and fills in your current IP address.
 12. Leave "Outbound rules" as they are (the default allows all outbound traffic, which is fine).
 13. Scroll to the bottom and click the orange "Create security group" button.
- ✓ **How you'll know it worked:** You're taken back to the Security Groups list, and you can see "meridian-ec2-sg" in the table.

WARNING

If you use mobile data or a VPN and it changes your IP address later, SSH access may stop working — you'd need to come back to this security group and update the SSH rule's source. This does not affect the website itself, only your ability to SSH in for debugging.

Step 3 — Create the database's security group (meridian-rds-sg)

1. Still on the Security Groups page, click "Create security group" again.
2. Security group name: meridian-rds-sg
3. Description: Allows the app server to reach the database
4. VPC: leave it as the same default VPC as before.
5. Under "Inbound rules", click "Add rule".
6. Set "Type" to "MYSQL/Aurora" (Port range fills in as 3306 automatically).

7. Click the "Source" box. Instead of picking "Anywhere" or typing an IP, start typing: meridian-ec2-sg — a matching security group will appear in the dropdown. Click it to select it.
8. Leave "Outbound rules" as default.
9. Scroll down and click "Create security group".

✓ **How you'll know it worked:** The Security Groups list now shows both meridian-ec2-sg and meridian-rds-sg.

WHY THIS MATTERS

By setting the database's Source to the security group (not an IP address), you're saying "allow anything that has meridian-ec2-sg attached to it" rather than "allow this one specific IP". This is the single most important trick for avoiding the connectivity problems you mentioned running into before — it keeps working even if the server restarts and gets a new IP address.

Take screenshot 01-security-groups.png now — see the Screenshots part later in this guide for the full list.

Part C — Create the Database

Step 4 — Launch the RDS MySQL database

1. In the top search bar, type: RDS, and click "RDS" in the results.
 2. In the left sidebar, click "Databases".
 3. Click the orange "Create database" button.
 4. Under "Choose a database creation method", select "Standard create".
 5. Under "Engine options", click "MySQL".
 6. Leave "Engine Version" at its default (a recent MySQL 8.0.x version).
 7. Under "Templates", click "Free tier". This automatically picks free-tier-safe settings for you.
 8. Under "Settings": in "DB instance identifier", type: meridian-goods-db
 9. Under "Master username", leave it as: admin
 10. Under "Credentials management", leave "Self managed" selected.
 11. In "Master password", type a password of your choosing (8+ characters, avoid the characters / @ " and spaces). Write this password down somewhere safe — you'll need it in Step 6.
 12. Re-type the same password in "Confirm master password".
 13. Under "Instance configuration", leave the default burstable free-tier-eligible class selected (e.g. db.t3.micro or db.t4g.micro — whichever is pre-selected under the Free tier template is correct, don't change it).
 14. Under "Storage", leave all defaults as-is.
 15. Under "Connectivity": for "Compute resource", choose "Don't connect to an EC2 compute resource" (you'll connect it yourself using the security group you already made).
 16. Still under Connectivity: "Virtual private cloud (VPC)" should show the default VPC — leave it.
 17. "Public access": click "No". This is important — it keeps the database from ever being reachable directly from the internet.
 18. "VPC security group": choose "Choose existing". A checklist of security groups appears — uncheck "default" if it's checked, and check the box next to meridian-rds-sg only.
 19. Leave "Availability Zone" as "No preference".
 20. Expand "Additional configuration" near the bottom of the page (click to expand it).
 21. In "Initial database name", type: meridian_goods — this is important, the app expects a database with this exact name to already exist.
 22. Leave every other setting under Additional configuration at its default.
 23. Scroll to the very bottom and click the orange "Create database" button.
- ✓ **How you'll know it worked:** You're taken to the Databases list, and meridian-goods-db appears with status "Creating".

Take screenshot 02-rds-creating.png now.

WARNING

Double-check "Public access" is set to No and the security group is meridian-rds-sg, not "default" — these are the two settings most commonly clicked past by accident, and they're the ones this whole design depends on for

security.

Step 5 — Wait for the database and copy its endpoint

1. The database takes roughly 5-10 minutes to finish setting up. You can leave this browser tab open — there's no need to refresh manually, but you can click the refresh icon on the page if you want to check progress.
 2. Wait until the "Status" column changes from "Creating" to "Available" (shown with a green icon).
 3. Once it says Available, click on the database name "meridian-goods-db" to open its details page.
 4. On the "Connectivity & security" tab, find the field labeled "Endpoint" — it looks like meridian-goods-db.xxxxxxxxxx.ap-south-1.rds.amazonaws.com
 5. Click the small copy icon next to it, or select and copy the text manually. Paste it somewhere you can find it again (a notes app, or a blank text file) — you'll need it in the next step.
- ✓ **How you'll know it worked:** Status shows "Available" with a green icon, and you have the endpoint copied somewhere safe.

Take screenshot 03-rds-available.png now, showing the endpoint visible.

Part D — Prepare Your Startup Script

Step 6 — Fill in your startup script

The project's GitHub repository includes a file called `infra/ec2-user-data.sh`. This script installs and starts the website automatically the moment the server turns on — you just need to fill in your real database details first.

1. Open `infra/ec2-user-data.sh` from the project (in GitHub, or from the copy on your computer) in any plain text editor — Notepad, TextEdit, or VS Code all work. Do not use Microsoft Word for this.
 2. Find the line that says: `DB_HOST="REPLACE_WITH_YOUR_RDS_ENDPOINT"` — replace the placeholder text (keeping the quotes) with the endpoint you copied in Step 5.
 3. Find the line that says: `DB_PASSWORD="REPLACE_WITH_YOUR_DB_PASSWORD"` — replace the placeholder with the master password you set in Step 4.
 4. Leave `DB_USER` and `DB_NAME` exactly as they are (`admin` and `meridian_goods`) — these already match what you set up in RDS.
 5. Save the file. Select all the text in the file (`Ctrl+A` / `Cmd+A`) and copy it (`Ctrl+C` / `Cmd+C`) — you'll paste this whole thing into AWS in the next step.
- ✓ **How you'll know it worked:** Your clipboard contains the entire script, starting with `#!/bin/bash`, with your real endpoint and password filled in (not the `REPLACE_` placeholders anymore).

WARNING

Make sure you copy the whole file, from the very first line to the very last. If you only copy part of it, the server won't set up correctly.

Part E — Launch the Server

Step 7 — Launch the EC2 instance

1. In the top search bar, type: EC2, and click "EC2".
 2. In the left sidebar, click "Instances".
 3. Click the orange "Launch instance" button.
 4. Under "Name and tags", in the "Name" box, type: meridian-goods-app
 5. Under "Application and OS Images (Amazon Machine Image)", make sure "Amazon Linux" is selected, and that the AMI shown is "Amazon Linux 2023 AMI" with a label saying "Free tier eligible". This is usually already selected by default — just confirm it.
 6. Under "Instance type", choose t2.micro (or t3.micro if that's the one marked "Free tier eligible" in your account/region — pick whichever one has that label).
 7. Under "Key pair (login)", click "Create new key pair". Name it: meridian-goods-key. Key pair type: RSA. Private key file format: .pem. Click "Create key pair" — a .pem file downloads to your computer automatically. Save it somewhere you'll remember (you likely won't need it, but keep it in case you ever want to debug via SSH).
 8. Under "Network settings", click the "Edit" button on the right side of that section.
 9. "VPC" should show the default VPC — leave it.
 10. "Subnet" — leave as "No preference".
 11. "Auto-assign public IP" — set this to "Enable". This is essential — without it, your website won't be reachable from the internet.
 12. "Firewall (security groups)" — select "Select existing security group".
 13. In the security group list that appears, check the box next to meridian-ec2-sg. If "default" is also checked, uncheck it.
 14. Leave "Configure storage" at its default (8 GiB, gp3) — this is within the free tier.
 15. Scroll down and click to expand "Advanced details".
 16. Scroll all the way to the bottom of Advanced details, to the box labeled "User data".
 17. Click inside that box and paste the script you copied in Step 6 (Ctrl+V / Cmd+V).
 18. Scroll back up. On the right-hand side "Summary" panel, confirm "Number of instances" is 1, then click the orange "Launch instance" button.
- ✓ **How you'll know it worked:** A green success banner appears saying your instance launch was successful, with a link to view your instance.

Take screenshot 04-ec2-launch-userdata.png before you click Launch — showing the pasted script in the Advanced details section (fine to keep the password visible for the screenshot, since only you'll see this folder, but blur it out if you're worried).

Step 8 — Wait for it to boot and get its public IP

1. Click "View all instances" (or go to EC2 → Instances from the sidebar).
2. Find meridian-goods-app in the list. Wait until "Instance state" shows "Running" (green) and "Status check" shows "2/2 checks passed" — this can take 2-3 minutes.
3. Click on the instance to open its details.

4. Find "Public IPv4 address" on the details panel and copy it — this is the address you'll use to visit your website.

✓ **How you'll know it worked:** Instance state: Running, Status check: 2/2 checks passed, and you have a public IP address copied, e.g. 13.234.56.78.

Take screenshot 05-ec2-running.png now.

IMPORTANT — GIVE IT A FEW MORE MINUTES

"Running" only means the server has turned on — it does NOT mean the website is ready yet. The script from Step 6 is still installing software and starting the app in the background. Wait an extra 3-4 minutes after the instance shows "Running" before testing in your browser.

Part F — Check Everything Works

Step 9 — Open the website

1. Open a new browser tab.
2. Type `http://` followed by the public IP address you copied, for example: `http://13.234.56.78`
3. Press Enter.

✓ **How you'll know it worked:** The Meridian Goods homepage loads, showing a grid of sample products like "Handwoven Cotton Throw" and "Cast Iron Skillet".

TIP

If nothing loads (the page just spins or times out), wait another few minutes and try again — see Part F's note above. If it still doesn't load after 10 minutes, check the Troubleshooting table.

Take screenshot 06-app-running.png now.

Step 10 — Check the health endpoint

1. In the same browser, go to: `http://<your-public-ip>/healthz` — for example `http://13.234.56.78/healthz`
- You should see this exact text on the page:

```
{"status": "ok", "db": "reachable"}
```

✓ **How you'll know it worked:** The page shows `"db": "reachable"` — this confirms the server can actually talk to your database, not just that the server itself is on.

Take screenshot 07-healthz-check.png now.

Step 11 — Test adding a product

1. Go back to the homepage (`http://<your-public-ip>/`).
2. Click "List a product" in the top-right of the page.
3. Fill in a name, description, price, and stock quantity for any product you like.
4. Click "Add to catalog".

✓ **How you'll know it worked:** You're taken back to the homepage and your new product appears in the grid — this confirms writes to the database are working, not just reads.

Take screenshot 08-add-product-test.png now, showing your product on the homepage.

Part G — Screenshots Checklist

Make sure you have all 8, saved into the project's screenshots/ folder with these exact filenames:

- 01-security-groups.png
- 02-rds-creating.png
- 03-rds-available.png
- 04-ec2-launch-userdata.png
- 05-ec2-running.png
- 06-app-running.png
- 07-healthz-check.png
- 08-add-product-test.png

Part H — Clean Up (Do This When You're Done)

WHY THIS STEP MATTERS

Free tier covers limited hours per month. If you leave the server and database running indefinitely, you risk being charged once you exceed those limits. Always delete both once you've finished testing and collected your screenshots.

Step 12 — Terminate the EC2 instance

1. Go to EC2 → Instances.
2. Check the box next to meridian-goods-app.
3. Click the "Instance state" button near the top, then choose "Terminate instance" from the dropdown.
4. Confirm by clicking "Terminate" in the popup.

✓ **How you'll know it worked:** "Instance state" changes to "Shutting-down" and then "Terminated" after a minute or two.

Step 13 — Delete the RDS database

1. Go to RDS → Databases.
2. Click on meridian-goods-db to open it, or check its box from the list.
3. Click the "Actions" button, then choose "Delete".
4. A confirmation panel opens. Uncheck "Create final snapshot" (you don't need one for this project).
5. Also uncheck "Retain automated backups" if shown.
6. Type delete me into the confirmation box exactly as instructed on screen.
7. Click the "Delete" button.

✓ **How you'll know it worked:** The database's status changes to "Deleting" and it disappears from the list after a few minutes.

Step 14 — (Optional) Delete the security groups

1. Go to EC2 → Security Groups.
2. Select meridian-ec2-sg and meridian-rds-sg (checkboxes on the left).
3. Click "Actions", then "Delete security groups", then confirm.

✓ **How you'll know it worked:** Neither security group appears in the list anymore. (If deletion fails saying it's still in use, wait a few more minutes for the EC2/RDS deletion above to fully finish, then try again.)

Troubleshooting — Common Problems and Exact Fixes

Symptom	Why it happens	Exact fix
Website doesn't load at all (times out)	Instance still booting, or the security group is missing the HTTP rule.	Wait a few more minutes first. Then check EC2 → Security Groups → meridian-ec2-sg → Inbound rules — confirm an HTTP rule exists with source 0.0.0.0/0.
Website loads but shows a server error	The app can't reach the database — usually a typo in the endpoint or password in your script.	Terminate the instance, re-check your copied script for the exact endpoint and password, and launch a new instance with the corrected script.
/healthz shows "db": "unreachable"	The RDS security group isn't correctly allowing the EC2 security group, or the database isn't Available yet.	Check RDS → your database → Connectivity & security → confirm meridian-rds-sg is attached. Check EC2 → Security Groups → meridian-rds-sg → Inbound rules → confirm the source is meridian-ec2-sg, not an IP or "default".
RDS status stuck on "Creating" for a long time	This is often just normal — free tier database creation can take up to 10-15 minutes.	Wait. If it's been over 20 minutes, refresh the page once; if still stuck, check the "Recent events" tab on the database's page for a specific error.
Can't find "Free tier eligible" instance type	You may be in a region where the label isn't shown the same way, or you scrolled past it.	t2.micro or t3.micro are both free-tier eligible in virtually every region — look for either one; the exact label wording can vary slightly by region.
Accidentally selected the "default" security group instead of your own	Easy to do — it's often pre-checked.	Uncheck "default" and check only meridian-ec2-sg (for the instance) or meridian-rds-sg (for the database).
Pasted user data script doesn't seem to have run	Either the whole file wasn't copied, or the placeholders weren't replaced before pasting.	Terminate the instance and launch a new one — user data only runs once, at first boot, so you can't re-run it on the same instance. Double check the full file was copied this time.
Region shows differently on EC2 vs RDS pages	You changed the region dropdown partway through the guide.	Always re-check the top-right corner before starting each Part. If they don't match, your resources can't find each other — recreate whichever one is in the wrong region.
"Create database" button is greyed out on the RDS page	A required field further up the form was left blank or invalid (commonly the password fields not matching).	Scroll back up and check every field with a red asterisk or red outline — usually the master password and its confirmation don't match.

How to check the server's setup log (advanced / optional)

If the website still isn't working after checking the table above, you can look at the server's own log of what happened during setup. This requires the .pem key file you downloaded in Step 7.

1. On Mac/Linux, open Terminal and run this once to fix the key file's permissions: `chmod 400 path/to/meridian-goods-key.pem`
2. Connect: `ssh -i path/to/meridian-goods-key.pem ec2-user@<your-public-ip>`
3. Once connected, run: `sudo cat /var/log/user-data.log`
4. Read from the bottom up — the most recent lines are usually where an error shows up, often in red or containing the word "error" or "Traceback".

TIP

On Windows, use PuTTY with the .ppk version of your key, or install the free "OpenSSH" feature in Windows

Settings to use the same ssh command shown above in PowerShell.

Final Submission Checklist

- ☐ Case study report (PDF) — includes your name, roll number, and GitHub link
- ☐ Source code — app/ folder pushed to GitHub
- ☐ AWS architecture diagram — diagrams/ folder pushed to GitHub
- ☐ Deployment files — infra/ec2-user-data.sh and infra/security-group-rules.md pushed to GitHub
- ☐ Deployment screenshots — all 8 images in screenshots/, pushed to GitHub
- ☐ README with setup/deploy/run/test instructions — pushed to GitHub
- ☐ GitHub repository is set to Public
- ☐ GitHub link appears in both the README and the case study report
- ☐ You terminated the EC2 instance and deleted the RDS database after finishing